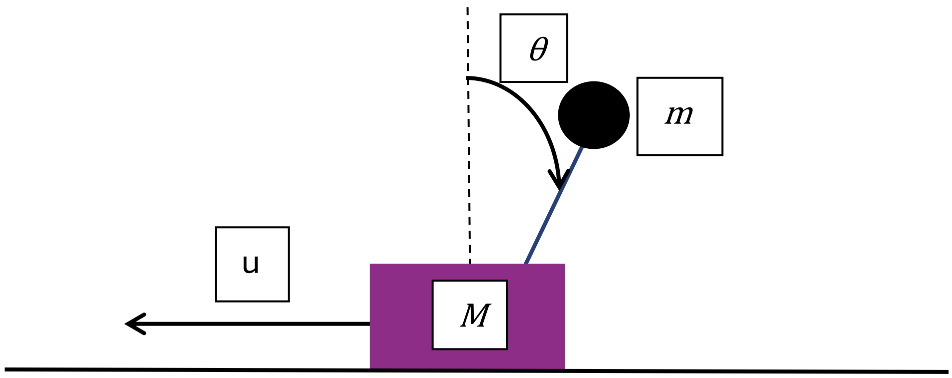
Inverted Pendulum on a cart



1. System Modeling
   1.1. Non-Linear Model

$$\ddot{\boldsymbol{x}} = -\frac{\boldsymbol{m\,g}}{\boldsymbol{M}}\,\boldsymbol{\theta} - \frac{\boldsymbol{d}}{\boldsymbol{M}}\dot{\boldsymbol{x}} + \frac{\boldsymbol{1}}{\boldsymbol{M}}\,\boldsymbol{u}\,,\ddot{\boldsymbol{\theta}} = \frac{\boldsymbol{g(M+m)}}{\boldsymbol{Ml}} + \frac{\boldsymbol{d}}{\boldsymbol{Ml}} - \frac{\boldsymbol{u}}{\boldsymbol{Ml}}$$

Where, $\boldsymbol{d}$ is cart damping and **u** is motor input torque (τ)

   1.2. Linear Model

$$\dot{\boldsymbol{x}} = \boldsymbol{A\,x} + \boldsymbol{B\,u}$$

$$\boldsymbol{y} = \boldsymbol{C\,x} + \boldsymbol{D\,u}$$

$$\dot{\boldsymbol{x}} = \begin{bmatrix}\dot{x}\\ \ddot{x}\\ \dot{\theta}\\ \ddot{\theta}\end{bmatrix}, \boldsymbol{x} = \begin{bmatrix}x\\ \dot{x}\\ \theta\\ \dot{\theta}\end{bmatrix}$$

$$\boldsymbol{A} = \begin{bmatrix}0 & 0 & 1 & 0\\ 0 & -\frac{d}{M} & -\frac{mg}{M} & 0\\ 0 & 0 & 0 & 1\\ 0 & \frac{d}{Ml} & g\left(\frac{m+M}{Ml}\right) & 0\end{bmatrix}, \boldsymbol{B} = \begin{bmatrix}0\\ \frac{1}{M}\\ 0\\ -\frac{1}{Ml}\end{bmatrix}$$

2. Controller
   2.1. Energy-Based Swing -up controller

$$E_{total} = \frac{1}{2}ml^2\dot{\theta}^2 - mgl\,cos(\theta)$$

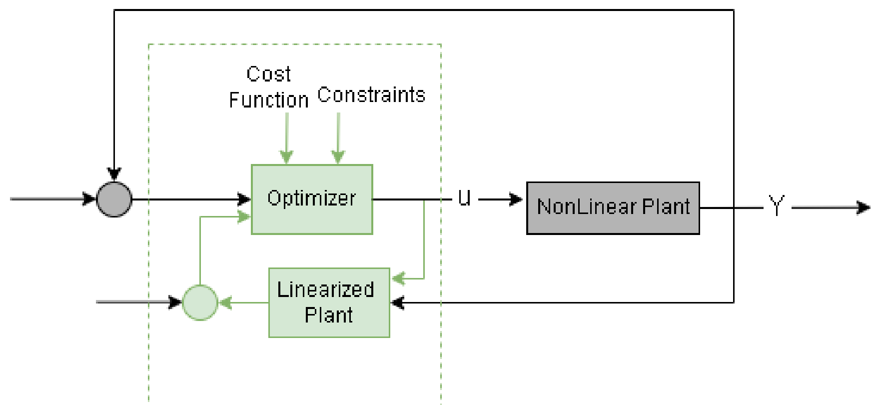
$$E_{UP} = mgl$$

$$E_{error} = E_{total} - E_{UP}$$

$$u_{swing} = -K_e(E - E_{UP})sign(\cos(\theta_1))$$

2.2. Up-right controller

2.2.1. Model predictive control



$$\min J = |\boldsymbol{y_t} - \boldsymbol{y}|$$

$$s.t. \qquad \dot{\boldsymbol{x}} = A\boldsymbol{x} + B\boldsymbol{u}$$

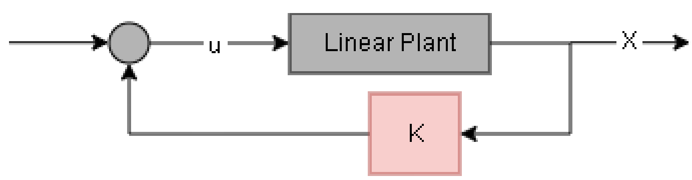
$$\boldsymbol{u_{low}} < \boldsymbol{u} < \boldsymbol{u_{high}} \qquad \boldsymbol{y_{t_{low}}} < \boldsymbol{y_t} < \boldsymbol{y_{t_{high}}}$$

2.2.2. Linear Quadratic Regulator

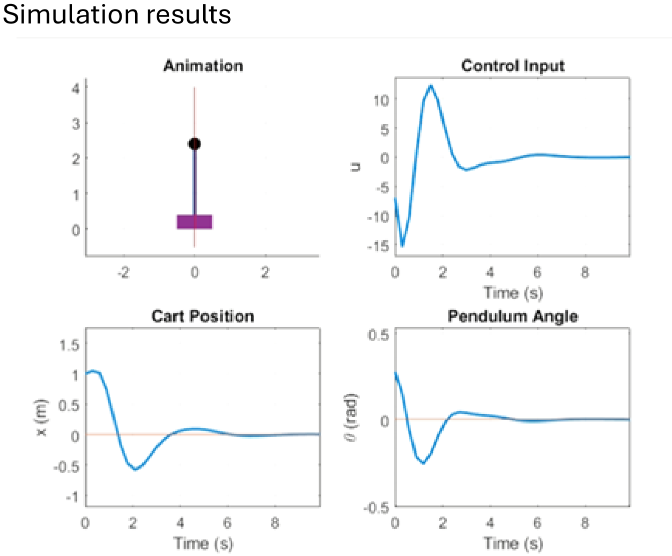
By minimizing cost function and solving Riccatti equation, K is derived

$$J = \int_0^{\infty} (x^T Q\, x + u^T R u) dt$$

$$\dot{\boldsymbol{x}} = \boldsymbol{A}\,\boldsymbol{x} + \boldsymbol{B}\,\boldsymbol{u}, \boldsymbol{u} = -\boldsymbol{K}\,\boldsymbol{x}$$



3. Simulation results



Figur.1 LQR contoller for small angle error

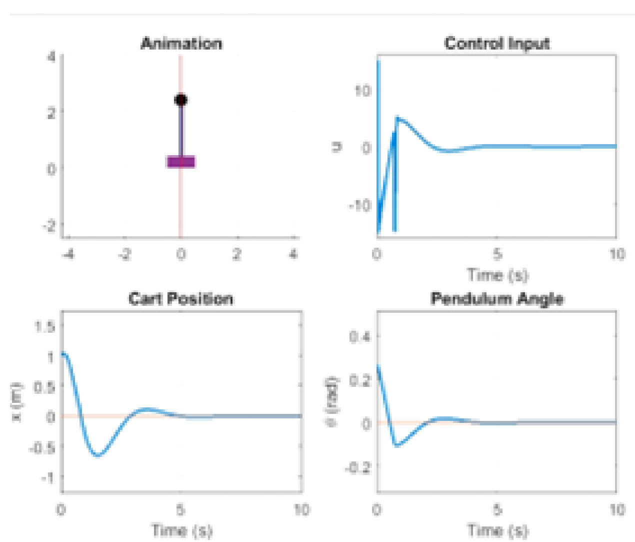


Figure.2 MPC controller for small angle error

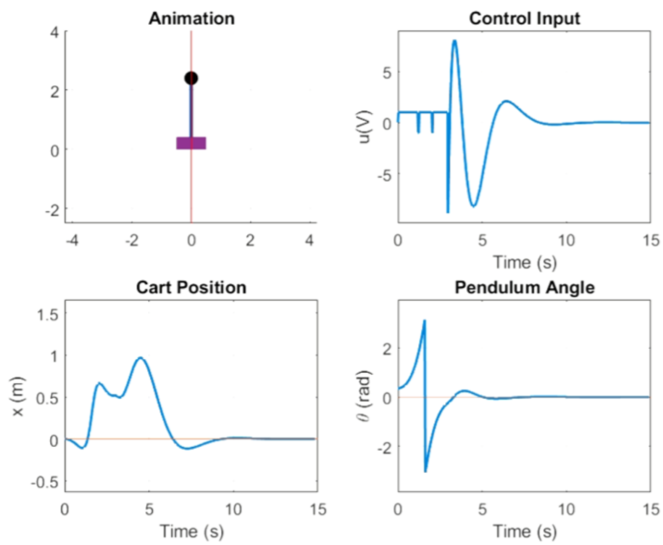


Figure.3 Swing-up with LQR for large angle error

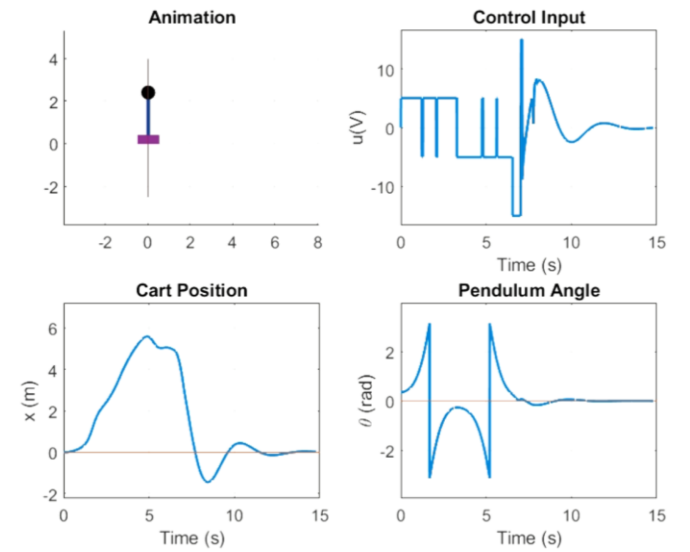


Figure.4 Swing-up with MPC for large angle error